

Project 1: Classification/Regression

In this project you will develop a classifier or regressor model for one of the following datasets. Note that some datasets have multiple aspects that can be predicted, you may choose from predicting the targets listed in the table. Training and testing data for each dataset can be found on GitHub. You may convert regression tasks into classification tasks by choosing a threshold, but explain justification for this.

Datasets

Name	Type	Details	Reference or Data Source
Lipophilicity	Regression	Predict log of octanol/water partition coefficient from SMILES	DeepChem/MolNet
Tox21 – SR_ARE	Classification	Predict if compound activates a stress response (antioxidant responsive element) that indicates possible toxicity	https://tripod.nih.gov/tox21/challenge/data.jsp
Bacterial membrane localization	Classification	Predict if bacterial proteins are membrane associated from protein sequence	https://db.psort.org/downloads

Model Requirements

You must write a model that contains the following aspects:

- Feature selection or dimensionality reduction, using at least one of the following techniques (Note: using Ridge/Lasso/Elastic Net does mitigate overfitting, but if you choose to use one of these techniques **you are still required to use an additional feature selection method**)
 - o Expert-chosen descriptors (if you choose this, you should explain your reasoning behind chosen descriptors using chemical or biological insight)
 - o Dimensionality reduction with PCA
 - o Statistical analysis of feature redundancy, predictive power/feature importance

- Features must be scaled if they do not fall within a standard range, if your features fall within a standard range (e.g. one hot encoding), you do not need to scale your features
- You must use GridSearchCV to optimize hyperparameters. You do not need to exhaustively test many combinations (so that your models can train in a reasonable amount of time) but you must test at least a few combinations
- Use one of the following models types: logistic regression, support vector machines, random forest or related method like XGBoost or Extremely randomized trees, Lasso Regression, Ridge Regression, Elastic Net Regression
- Models must show reasonable accuracy on the test set

Report Format/Requirements

Reports should be written in sentence format (no bullet points) and follow the format below:

Abstract

Describe the goal of what you are trying to predict, the significance of this problem, summarize your approach and the results (refer to reference for dataset source if you need more information on significance) **between 150-300 words**

Methods

Explain your featurization approach, and model design. Justify your choice of feature selection method and use or lack of use of feature scaling. **No length requirement but must contain all information needed so that someone familiar with AI could reproduce what you have done**

Results + Discussion

Discuss results of GridSearchCV and what the optimal hyperparameters were

Discuss accuracy of model on training and test data:

- Should include at least one graphic (ROC or PRC curve for classification, true vs predicted for regression)
- Discuss severity of overfitting
- Discuss if model would be useful in its domain based on test accuracy

At least 400 words

Conclusion

Summarize results, discuss possible future directions or areas for improving the model
(150-300 words)

Generative AI disclosure

Disclose if/how you have used generative AI for writing code and this report. If you used GenAI, include a copy of the chat or a list of prompts used. If you demonstrate that you have fully thought through the steps needed to complete the assignment through your prompts, you will get full credit on the metacognition item on the rubric. If the chat demonstrates that the AI filled in significant gaps of understanding, points will be deducted (see below for more details). Full credit is automatically given if you did not use GenAI.

Reports must be written in your own words and you may not use generative AI to generate the report. Generative AI for report writing is only permitted for understanding concepts or editing for clarity or grammar. **You must also disclose use of Generative AI for these tasks.**

Examples of GenAI use that will get full credit:

- You write out all steps needed for the code in natural language and ask the AI to translate it into code for you, example of sufficient detail for code that reads a fasta file and counts the hydrophobic amino acids in a protein:

Write a python script that does the following: the script should read a fasta file using the Biopython library and make a dictionary of all protein sequences, where the headers are the key and the values are the sequences. The script should then loop through all keys. In this loop, the number of hydrophobic amino acids in each sequence should be counted. This count should happen in an additional nested loop that iterates through the amino acids in the sequence and adds to the count if that amino acid is found in the list of hydrophobic amino acids. The hydrophobic amino acids have the following one letter codes: G, P, F, A, I, L, V. The count of hydrophobic amino acids should be stored in a second dictionary where the key is the header and the value is the hydrophobic amino acid count.

- You write the code yourself but occasionally use GenAI to remind yourself of how to do certain things in python, these should be restricted to answers that would be less than five lines of code. For example, you could ask how to sort a dictionary in python.
- You write a near complete version of the code yourself, then use GenAI to help debug issues with the code or further improve model performance. This should

involve more than just cutting and pasting your code into the prompt and asking “How do I fix this?”, your prompt should include a description of the error or issue and your ideas about what might cause the error and you should demonstrate through your dialog with the AI that you have thought about what might have gone wrong with your code.

Combinations of the above will also be accepted.

You are welcome to show me how you are using GenAI before the assignment is due to get my assessment of how many points you would get for that usage.

Rubric

Item	0	1	2	3	4	5
Abstract						Abstract is required length and contains required information, summary is accurate
Methods		Both featurization and methods explanation insufficient		Either featurization justification or methods explanation is lacking		Featurization approach is explained and justified, methods explained such that they can be repeated
Results						Results meet required length, contain required discussion, discussion well reasoned
Figure(s)						Axes labeled and it is clear

						what figure displays
Conclusion						Conclusion is required length and contains required discussion, discussion well reasoned
Code	Code is nonfunctional or not relevant to report	Code differs significantly from report			Comments lacking	Code is runnable and does what is described in report, code is well commented
Demonstration of Metacognition	Code was generated using assignment text or slight variations on assignment	Description of what code should do in prompt is cursory and GenAI filled in most details			Prompts demonstrate significant thought about assignment but AI was used to fill in several gaps	No GenAI used OR GenAI used but chat demonstrates full understanding of process needed to complete assignment